

Introduction

We know that the greedy algorithms make decisions step-by-step, choosing the locally optimal solution at each step. These algorithms typically run fast, often in linear or linearithmic time (e.g., $O(n)$, $O(n \log n)$), however do not explore multiple combinations or paths like backtracking or dynamic programming.

The knapsack problem is a popular optimization problem in computer science and operations research. It involves choosing items with values and weights to maximize the total value without exceeding a weight limit. This handout focuses on:

- Using the **Greedy Approach** to solve both 0/1 and Fractional Knapsack problems.
- Understanding why greedy works for one and fails for the other.

1 Problem Setting

You are given:

- A knapsack with maximum capacity W .
- n items, each with:
 - Value v_i
 - Weight w_i

2 Greedy Strategy

The greedy idea is to pick items with the highest **value per unit weight**:

$$\text{Ratio } r_i = \frac{v_i}{w_i}$$

Sort the items in decreasing order of r_i , and then pick them until the knapsack is full.

3 Fractional Knapsack (Greedy Works!)

What's allowed: You can take **fractions** of items.

Algorithm:

1. Compute $r_i = \frac{v_i}{w_i}$ for each item.
2. Sort items by decreasing r_i .
3. Take full items while capacity allows.
4. If only part of the next item fits, take the fraction.

Example:

Item	Value	Weight	v/w
1	60	10	6.0
2	100	20	5.0
3	120	30	4.0

Knapsack Capacity = 50

Steps:

- Take all of item 1 $\Rightarrow$ 10 weight, 60 value
- Take all of item 2 $\Rightarrow$ 20 weight, 100 value
- Take 20/30 of item 3 $\Rightarrow$ 20 weight, 80 value

Total Value = $60 + 100 + 80 =$ 240

Time Complexity: $O(n \log n)$ due to sorting

4 0/1 Knapsack (Greedy Fails!)

What's allowed: You can take either the whole item or none.

Let's try the same greedy approach and see how it fails.

Item	Value	Weight	v/w
1	60	10	6.0
2	100	20	5.0
3	120	30	4.0

Knapsack Capacity = 50 Greedy (value/weight):

- Take item 1 (10 weight, 60 value)
- Take item 2 (20 weight, 100 value)
- Item 3 does not fit

Total value: 160

But optimal choice: Items 2 and 3 (weight 50) gives 220, which is better.

Conclusion: Greedy doesn't always give optimal result in 0/1 knapsack!

5 Why Greedy Works for Fractional Knapsack but Not 0/1

Key Differences: Greedy in Knapsack Variants

- **Fractional Knapsack:**
 - You *can take fractions* of items.
 - Greedy works optimally by picking items with highest **value/weight**.
- **0/1 Knapsack:**
 - You must *take the whole item or none*.
 - Greedy may fail — the best ratio doesn't guarantee the best total value.

Why? Once you greedily pick a heavy item, you may run out of space for a better *combination* of lighter items with higher total value.

Analogy: Backpack at a Fruit Market

Imagine you're at a fruit market with a backpack that can carry up to **10 kg**. You are presented with the following choices:

Fruit	Weight (kg)	Value (\$)
Watermelon	10	60
5 Apples	5	50
5 Bananas	5	50

Greedy choice (by highest value):

- Picks the **Watermelon**: uses full 10 kg, total value = \$60.

Better alternative:

- Pick **5 Apples + 5 Bananas**: uses full 10 kg, total value = \$100.

Lesson: Greedy takes the best item *now*, but misses out on the best *combination*. *A loose inspiration to the Cloud Computing paradigm!*

Mental Model

- **Fractional Knapsack** is like filling a water bottle with honey — you can pour any amount.
- **0/1 Knapsack** is like packing books in a bag — take all or none.